SWITCHING: PACKET AND CIRCUIT SWITCHING

Introduction

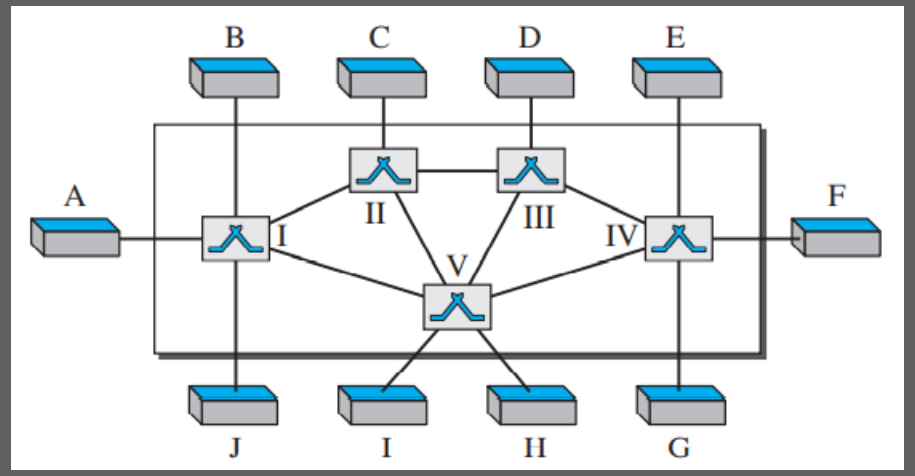
- A network is a set of connected devices.
- Multiple devices - we have the problem of how to connect them to make one-to-one communication possible.
- One solution is to make a point-to-point connection between each pair of devices (a mesh topology) or between a central device and every other device (a star topology).
- These methods - impractical and wasteful when applied to very large networks.
- The number and length of the links require too much infrastructure to be cost-efficient, and the majority of those links would be idle most of the time.

Contd…

- Other topologies employing multipoint connections, such as a bus, are ruled out because the distances between devices and the total number of devices increase beyond the capacities of the media and equipment.
- A better solution is switching. A switched network consists of a series of interlinked nodes, called **switches**.
- Switches are devices capable of creating temporary connections between two or more devices linked to the switch.
- In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing.

SWITCHED NETWORK

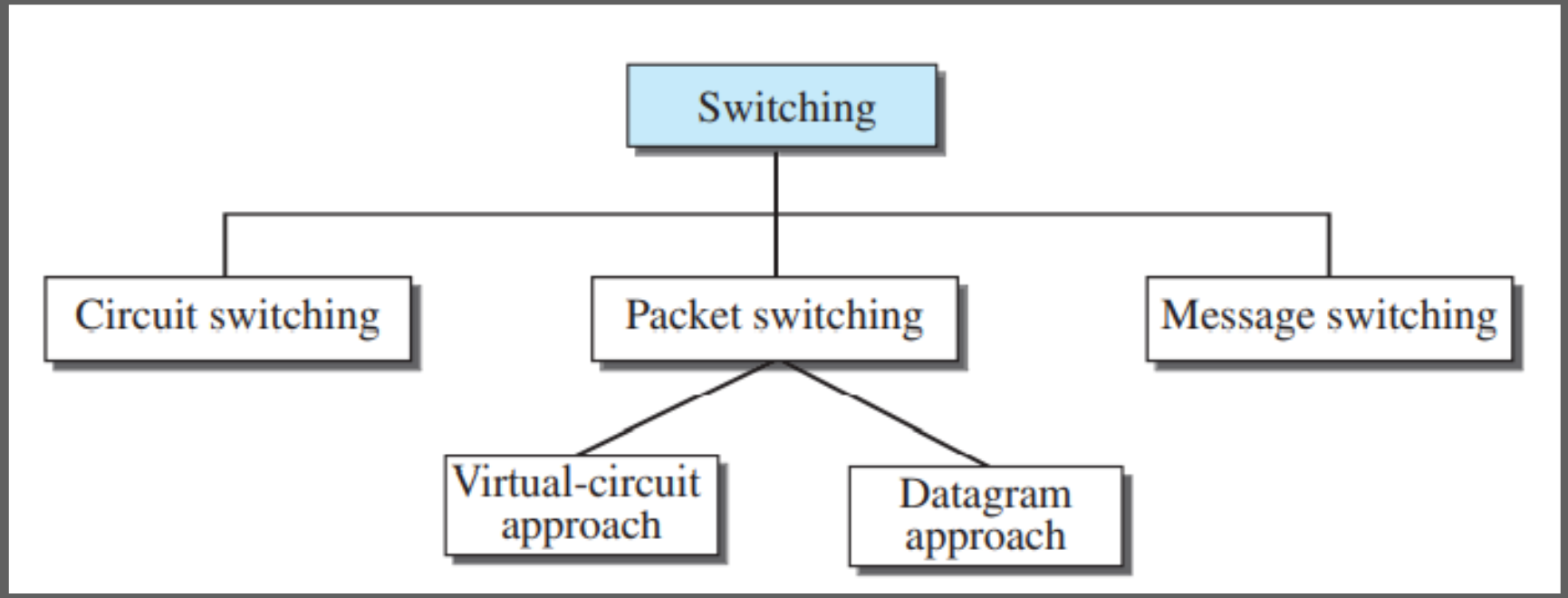
- The end systems (communicating devices) are labeled A, B, C, D, and so on, and the switches are labeled I, II, III, IV, and V.
- Each switch is connected to multiple links



THREE METHODS OF SWITCHING

- Three methods of switching
  - **Circuit switching**
  - **Packet switching**
  - **Message switching**
- Packet switching can further be divided into two subcategories **virtual circuit approach** and **datagram approach**

Taxonomy Of Switched Networks



SWITCHING AND TCP/IP LAYERS

- Switching can happen at several layers of the TCP/IP protocol suite.

Switching at Physical Layer:

- At the physical layer - circuit switching.
- There are no packets exchanged at the physical layer.
- The switches at the physical layer allow signals to travel in one path or another.

CONTD…

Switching at Data-Link Layer:

- At the data-link layer - packet switching.
- Packet - frames or cells.
- Packet switching at the data-link layer - Use virtual-circuit approach.

Switching at Network Layer

- At the network layer - packet switching.
- Virtual-circuit approach or a datagram approach can be used.

CONTD…

Switching at Application Layer :

- At the application layer - message switching.
- The communication at the application layer occurs by exchanging messages.
- Communication using e-mail is a kind of message-switched communication, network can be called a message-switched network.

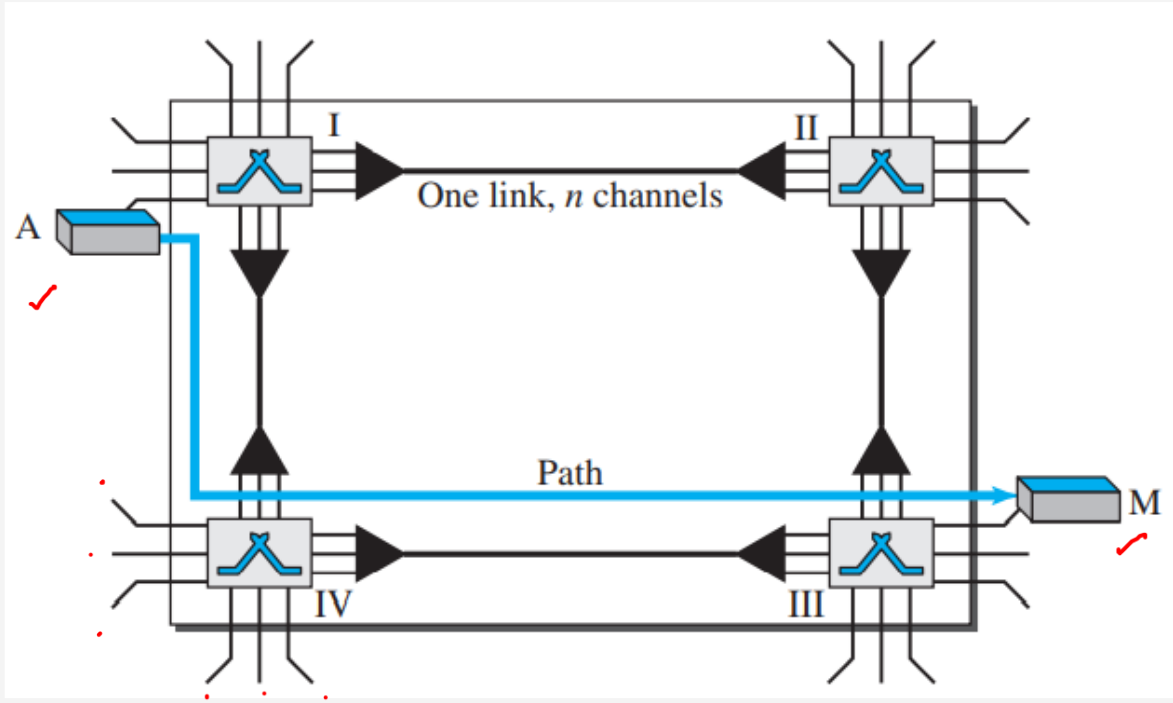
CIRCUIT-SWITCHED NETWORKS

INTRODUCTION

- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links.
- Each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.

A circuit-switched network is made of a set of switches connected by physical links, in which each link is divided into n channels

A TRIVIAL CIRCUIT-SWITCHED NETWORK



A trivial circuit-switched network with four switches and four links. Each link is divided into n (n is 3 in the figure) channels by using FDM or TDM.

CONTD…

- The end systems, such as computers or telephones, are directly connected to a switch.
- When end system A needs to communicate with end system M, system A needs to request a connection to M that must be accepted by all switches as well as by M itself.
- This is called the **setup phase**; a circuit (channel) is reserved on each link, and the combination of circuits or channels defines the dedicated path.
- After the dedicated path made of connected circuits (channels) is established, the **data-transfer phase** take place. After all data have been transferred, the circuits are torn down.

CONTD…

- Circuit switching takes place at the physical layer.
- Before starting communication, the stations must make a reservation for the resources to be used during the communication.
- These resources, such as channels (bandwidth in FDM and time slots in TDM), switch buffers, switch processing time, and switch input/output ports, must remain dedicated during the entire duration of data transfer until the teardown phase.
- Data transferred between the two stations are not packetized (physical layer transfer of the signal).

CONTD…

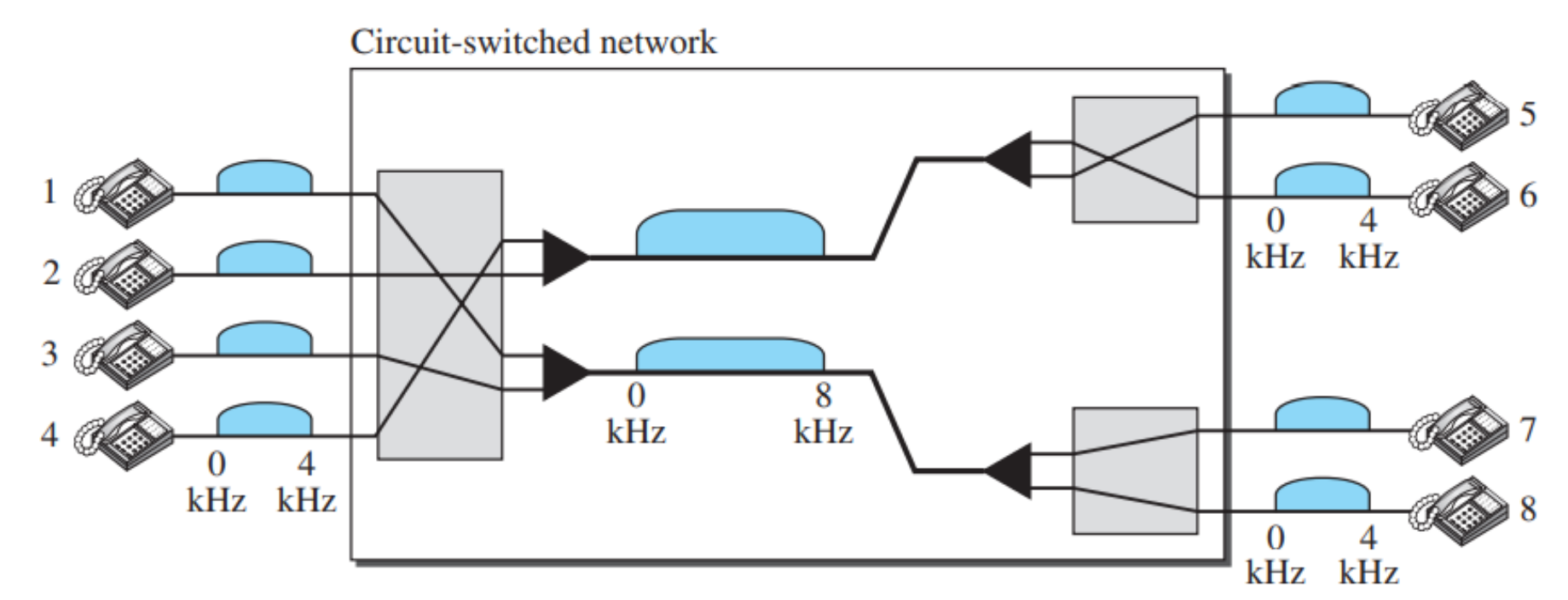
- The data are a continuous flow sent by the source station and received by the destination station, although there may be periods of silence.
- There is no addressing involved during data transfer.
- The switches route the databased on their occupied band (FDM) or time slot (TDM).

In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.

CONTD…

- As a trivial example, let us use a circuit-switched network to connect eight telephones in a small area.
- Communication is through 4-kHz voice channels.
- Each link uses FDM to connect a maximum of two voice channels. The bandwidth of each link is then 8 kHz.
- Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6.
- The switch controls the connections.

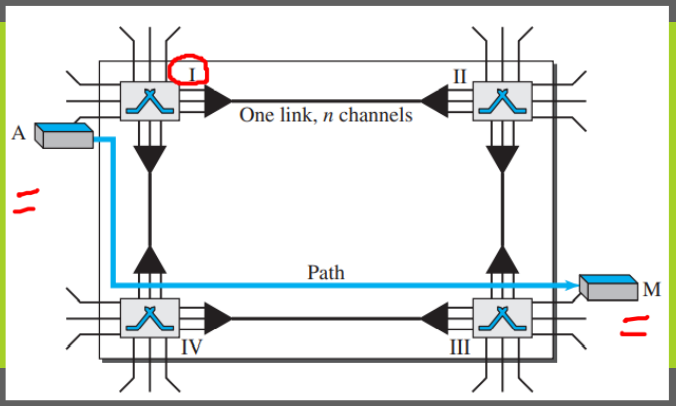
CONTD…


Circuit-switched network
1
2
3
4
0
kHz
4
kHz
0
kHz
8
kHz
0
kHz
4
kHz
5
6
7
8
0
kHz
4
kHz

THREE PHASES

- The actual communication in a circuit-switched network requires three phases:
  - Connection setup
  - Data transfer
  - Connection teardown
- Setup Phase Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established.
- The end systems are normally connected through dedicated lines to the switches, so connection setup means creating dedicated channels between the switches.

CONTD…



- For example, when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I.
- Switch I finds a channel between itself and switch IV that can be dedicated for this purpose.
- Switch I then sends the request to switch IV, which finds a dedicated channel between itself and switch III.
- Switch III informs system M of system A's intention at this time.

CONTD…

- In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A.
- Only after system A receives this acknowledgment is the connection established. Note that end-to-end addressing is required for creating a connection between the two end systems.
- These can be, for example, the addresses of the computers assigned by the administrator in a TDM network, or telephone numbers in an FDM network.

CONTD…

Data-Transfer Phase:

- After the establishment of the dedicated circuit (channels), the two parties can transfer data.

Teardown Phase:

- When one of the parties needs to disconnect, a signal is sent to each switch to release the resources

EFFICIENCY

- Circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection.
- In a telephone network, people normally terminate the communication when they have finished their conversation.
- In computer networks, a computer can be connected to another computer even if there is no activity for a long time.

DELAY

- Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal.
- During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.
- There is no waiting time at each switch.
- The total delay is due to the time needed to create the connection, transfer data, and disconnect the circuit.

CONTD…

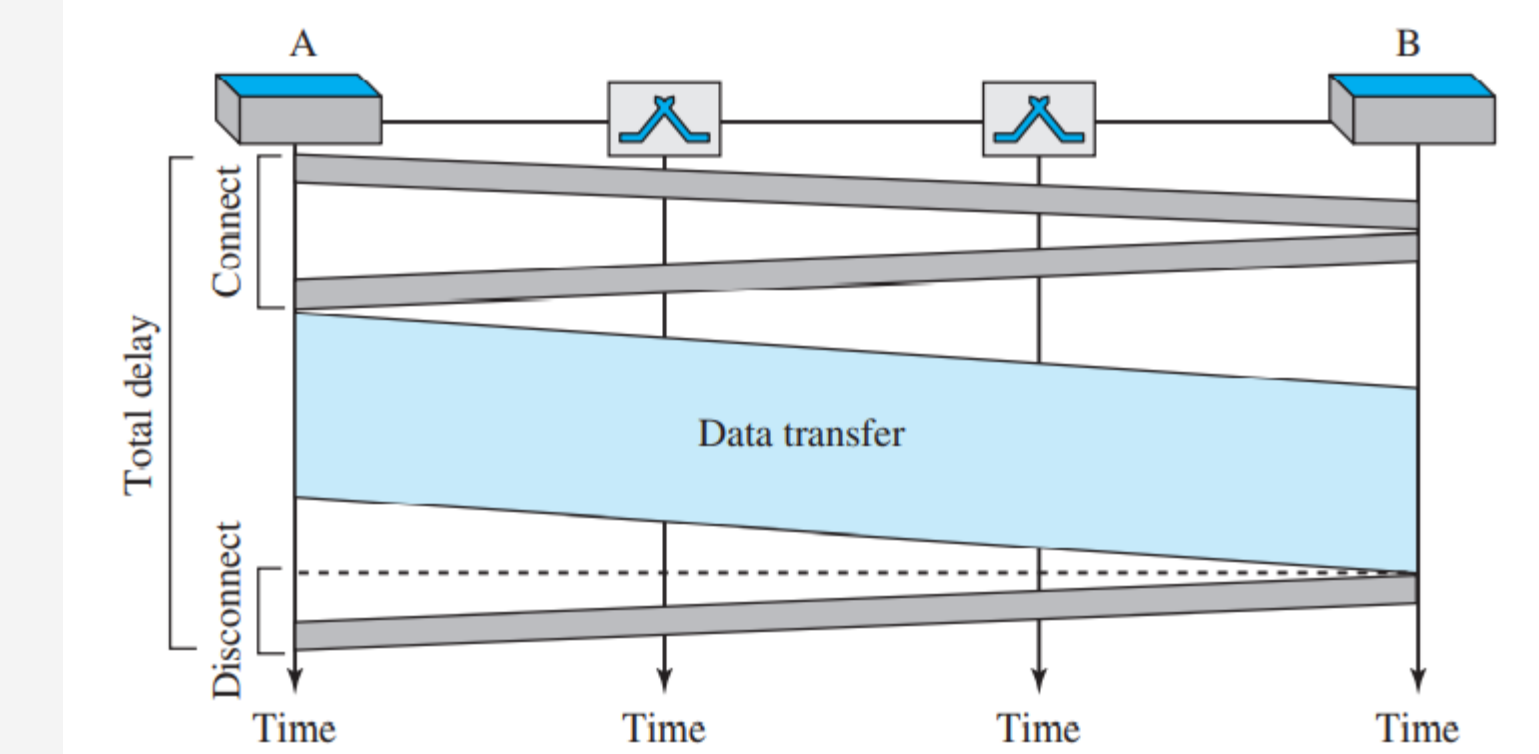
The delay caused by the setup is the sum of four parts:

- The propagation time of the source computer request (slope of the first gray box)
- The request signal transfer time (height of the first gray box)
- The propagation time of the acknowledgment from the destination computer (slope of the second gray box)
- The signal transfer time of the acknowledgment (height of the second gray box)

CONTD…

- The delay due to data transfer is the sum of two parts:
  - The propagation time (slope of the colored box)
  - Data transfer time (height of the colored box), which can be very long.
- The third box shows the time needed to tear down the circuit.

DELAY IN A CIRCUIT-SWITCHED NETWORK



PACKET SWITCHING

INTRODUCTION

- In data communications, we need to send messages from one end system to another.
- If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size.
- The size of the packet is determined by the network and the governing protocol. In packet switching, there is no resource allocation for a packet.
- This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet.
- Resources are allocated on demand.

CONTD…

- The allocation is done on a first come, first-served basis.
- When a switch receives a packet, no matter what the source or destination is, the packet must wait if there are other packets being processed.

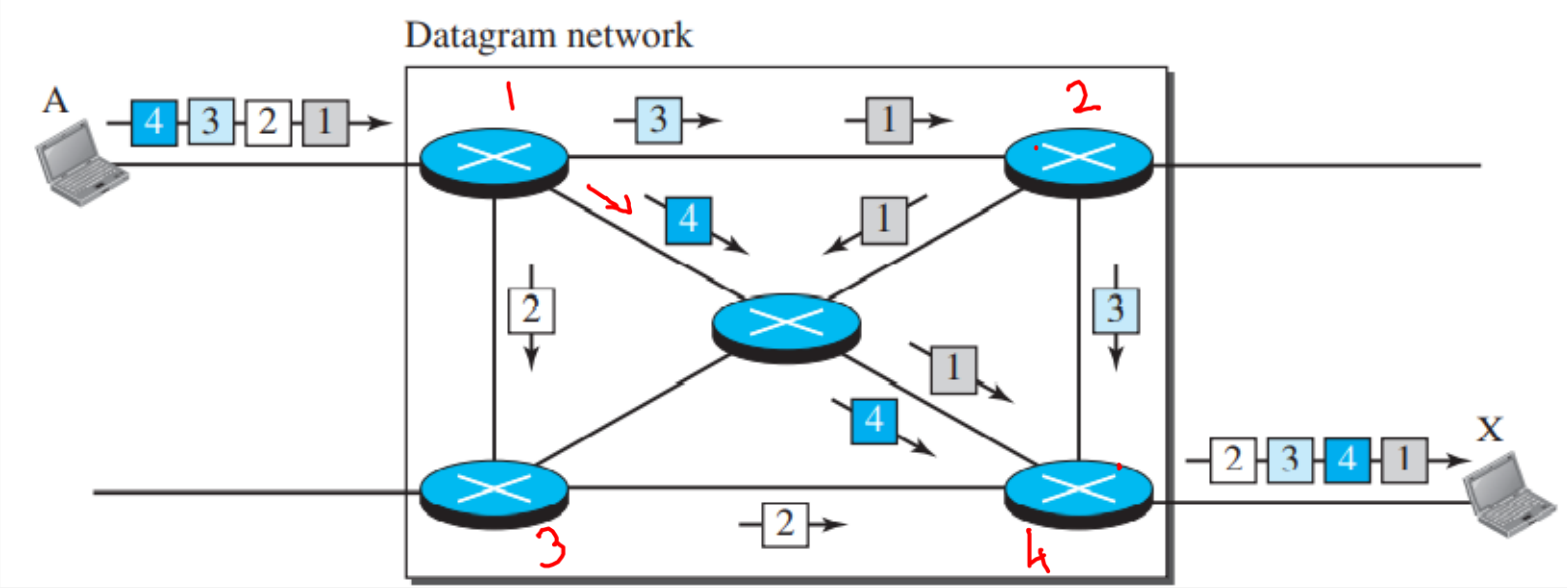
In a packet-switched network, there is no resource reservation; resources are allocated on demand.

- Two types of packet-switched networks:
- Datagram networks
- Virtual circuit networks

DATAGRAM NETWORKS

- In a datagram network, each packet is treated independently of all others.
- Even if a packet is part of a multipacket transmission, the network treats it as though it existed alone.
- Packets in this approach are referred to as datagrams.
- Datagram switching is normally done at the network layer.
- The datagram approach is used to deliver four packets from station A to station X.
- The switches in a datagram network are traditionally referred to as routers.

A DATAGRAM NETWORK WITH FOUR SWITCHES (ROUTERS)



CONTD…

- All four packets (or datagrams) belong to the same message, but may travel different paths to reach their destination.
- This is so because the links may be involved in carrying packets from other sources and do not have the necessary bandwidth available to carry all the packets from A to X.
- This approach can cause the datagrams of a transmission to arrive at their destination out of order with different delays between the packets.

CONTD..

- Packets may also be lost or dropped because of a lack of resources.
- In most protocols, it is the responsibility of an upper-layer protocol to reorder the datagrams or ask for lost datagrams before passing them on to the application.

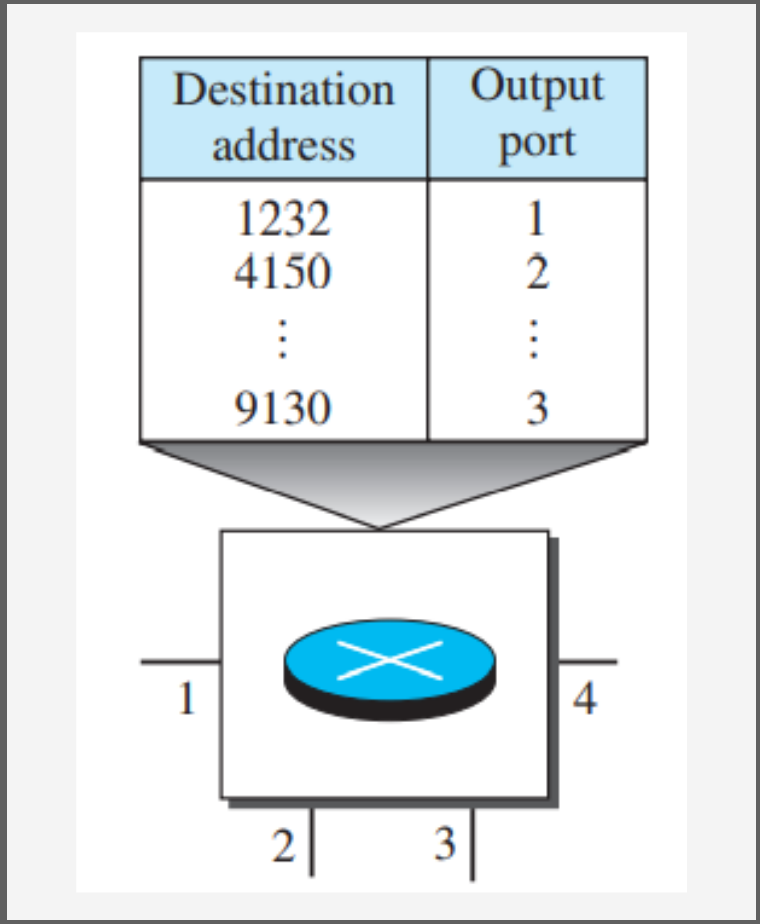
CONTD…

- The datagram networks - connectionless networks.
- The term connectionless - the switch (packet switch) does not keep information about the connection state.
- There are no setup or teardown phases.
- Each packet is treated the same by a switch regardless of its source or destination.

ROUTING TABLE

- If there are no setup or teardown phases,
- In this type of network, each switch (or packet switch) has a routing table which is based on the destination address.
- The routing tables are dynamic and are updated periodically.
- The destination addresses and the corresponding forwarding output ports are recorded in the tables.
- This is different from the table of a circuit switched network (discussed later) in which each entry is created when the setup phase is completed and deleted when the teardown phase is over.

ROUTING TABLE



A switch in a datagram network uses a routing table that is based on the destination address.

DESTINATION ADDRESS

- Every packet in a datagram network carries a header that contains, among other information, the destination address of the packet.
- When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded.
- This address, unlike the address in a virtual-circuit network, remains the same during the entire journey of the packet.

CONTD…

The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

EFFICIENCY

- The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred.
- If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources can be reallocated during these minutes for other packets from other sources.

DELAY

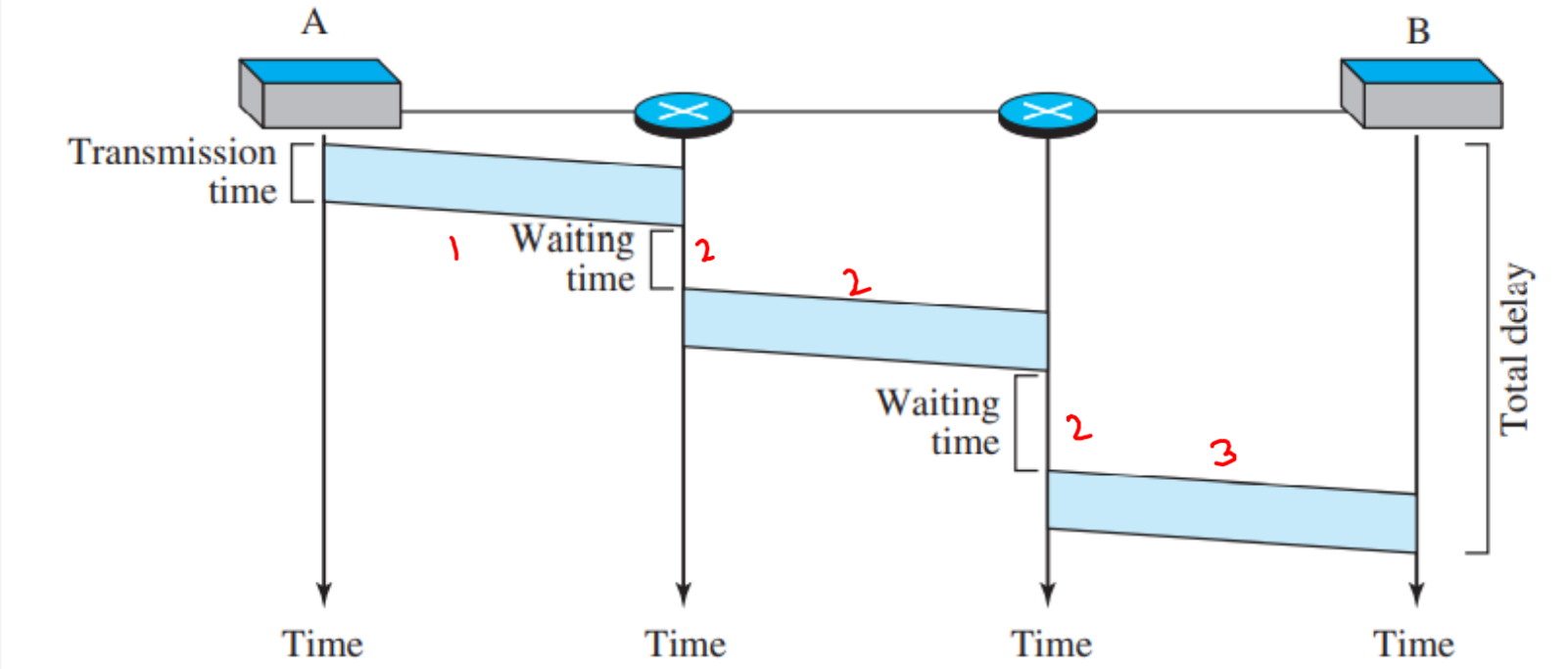
- There may be greater delay in a datagram network than in a virtual-circuit network.
- Although there are no setup and teardown phases, each packet may experience a **wait at a switch** before it is forwarded.
- In addition, since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.

CONTD…

- The packet travels through two switches.
- There are three transmission times (3T), three propagation delays (slopes 3τ of the lines), and two waiting times ($w_1 + w_2$).
- We ignore the processing time in each switch.
- The total delay is

$$\textbf{Total delay} = 3T + 3\tau + w_1 + w_2$$

DELAY IN A DATAGRAM NETWORK



VIRTUAL-CIRCUIT NETWORKS

VIRTUAL-CIRCUIT NETWORKS

- A virtual-circuit network is a cross between a circuit-switched network and a datagram network.

Some characteristics of both

1. As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.

2. Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network

CONTD…

3. As in a datagram network, data are packetized and each packet carries an address in the header.

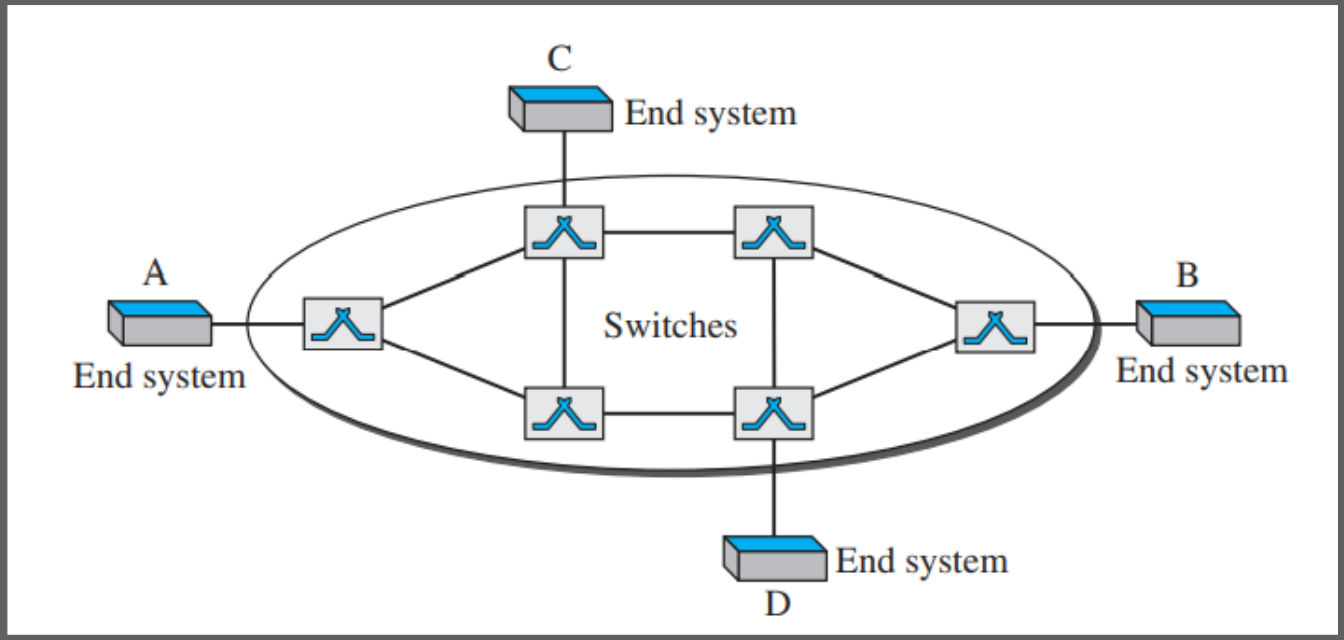
- The address in the header has local jurisdiction (it defines what the next switch should be and the channel on which the packet is being carried), not end-to-end jurisdiction.

4. As in a circuit-switched network, all packets follow the same path established during the connection.

5. A **virtual-circuit network - data-link layer,** while a circuit-switched network - physical layer and a datagram network - network layer.

VIRTUAL-CIRCUIT NETWORK

- The network has switches that allow traffic from sources to destinations.
- A source or destination can be a computer, packet switch, bridge, or any other device that connects other networks.



ADDRESSING

- Two types of addressing: **global and local** (virtual-circuit identifier).

Global Addressing:

- A source or a destination needs to have a global address—an address that can be unique in the scope of the network or an international network.
- A global address in virtual-circuit networks is used only to create a virtual-circuit identifier.

VIRTUAL-CIRCUIT IDENTIFIER

- The identifier used for data transfer is called the **virtual-circuit identifier (VCI) or the label**.
- A VCI is a small number that has only switch scope; it is used by a frame between two switches.
- When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI.
- The VCI in a data frame changes from one switch to another.

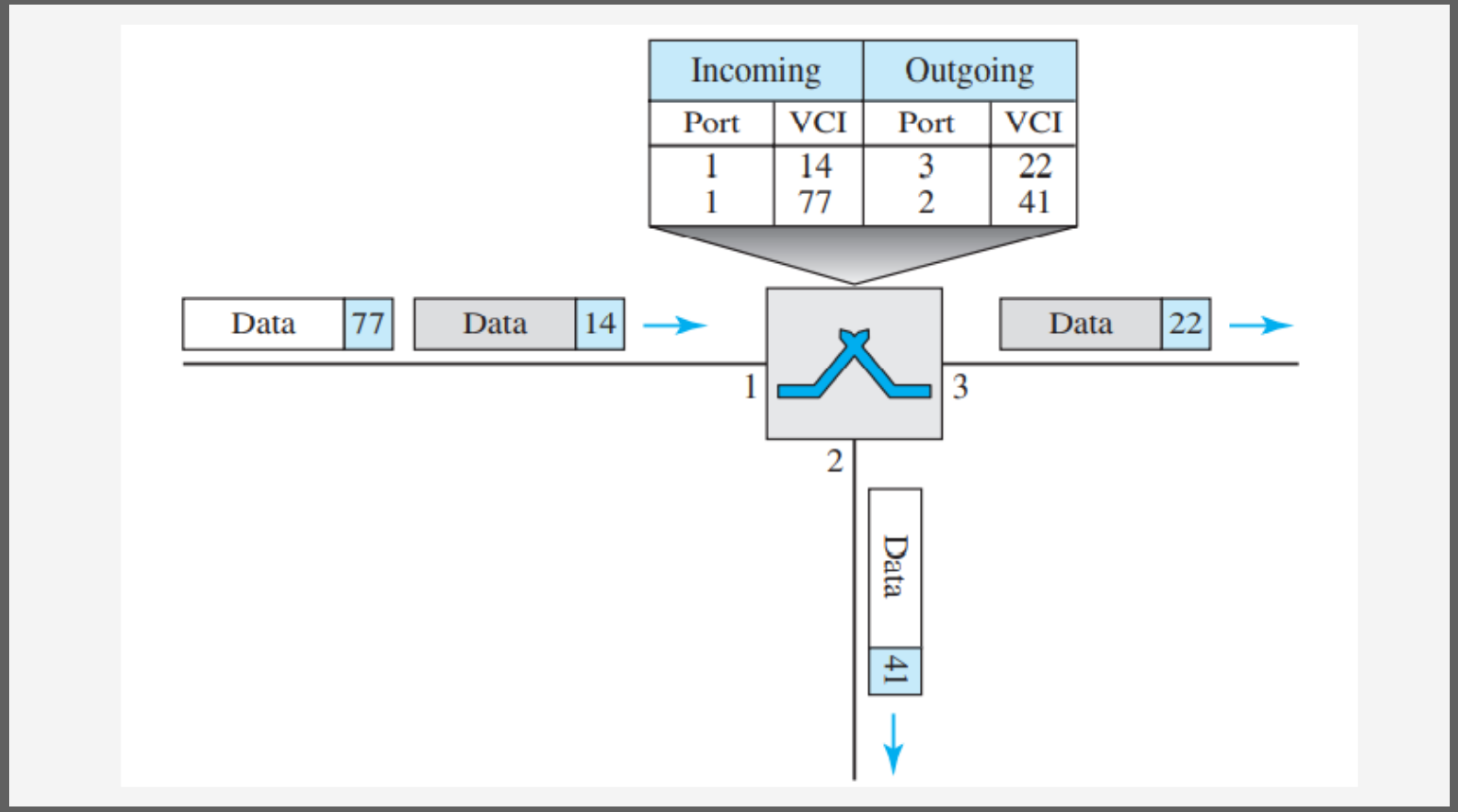
THREE PHASES

- Setup, data transfer, and teardown Phrase.
- In the **setup phase**, the source and destination use their global addresses to help switches make table entries for the connection.
- In the **teardown phase**, the source and destination inform the switches to delete the corresponding entry.
- **Data transfer** occurs between these two phases.

DATA-TRANSFER PHASE

- To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit.
- The table has four columns - four pieces of information for each virtual circuit that is already set up.

SWITCH AND TABLES IN A VIRTUAL-CIRCUIT NETWORK



CONTD…

- A frame arriving at port 1 with a VCI of 14.
- When the frame arrives, the switch looks in its table to find port 1 and a VCI of 14.
- When it is found, the switch knows to change the VCI to 22 and send out the frame from port 3.
- A frame from source A reaches destination B and how its VCI changes during the trip.
- Each switch changes the VCI and routes the frame.

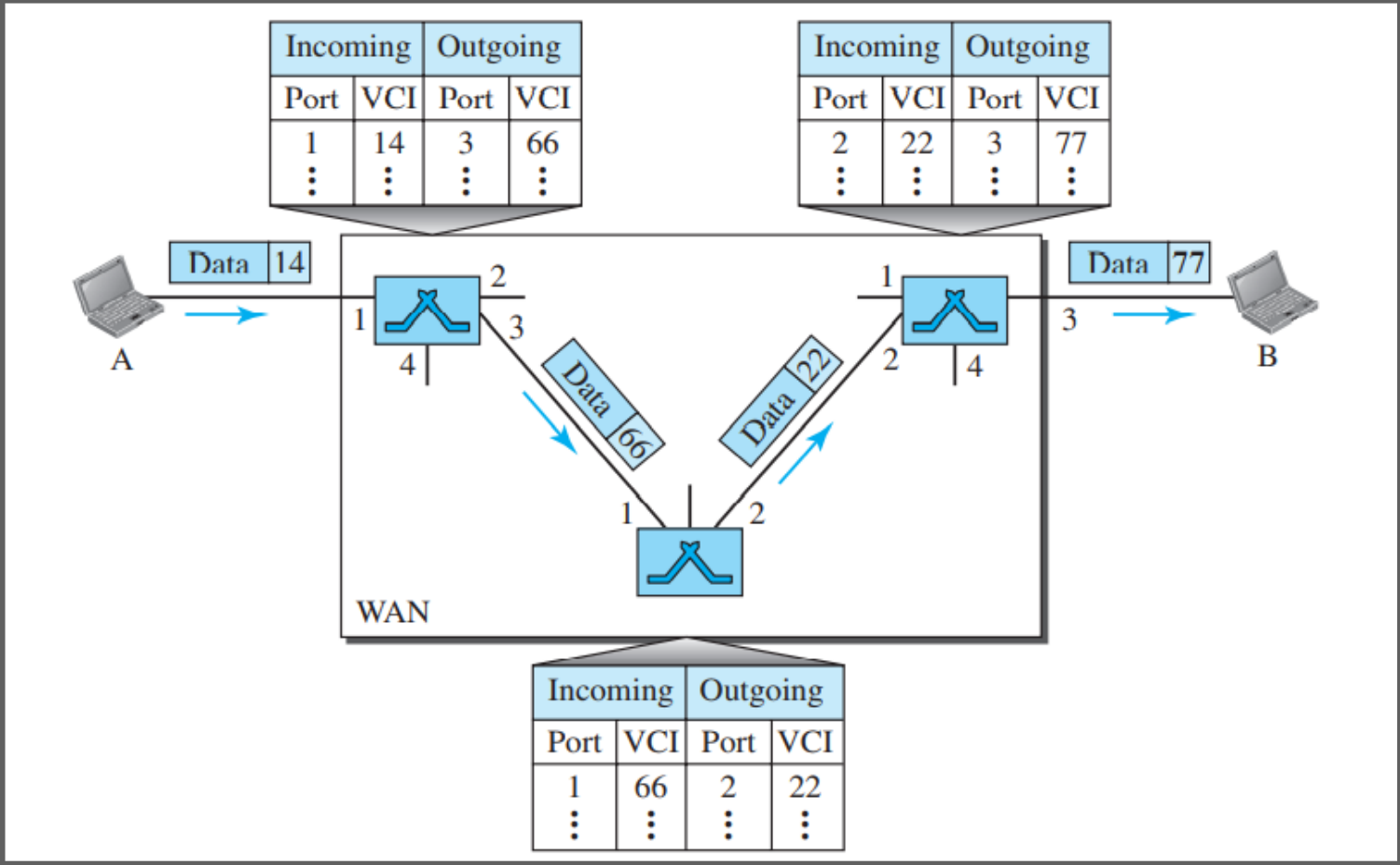
CONTD…

- The data-transfer phase is active until the source sends all its frames to the destination.
- The procedure at the switch is the same for each frame of a message.
- The process creates a virtual circuit, not a real circuit, between the source and destination.

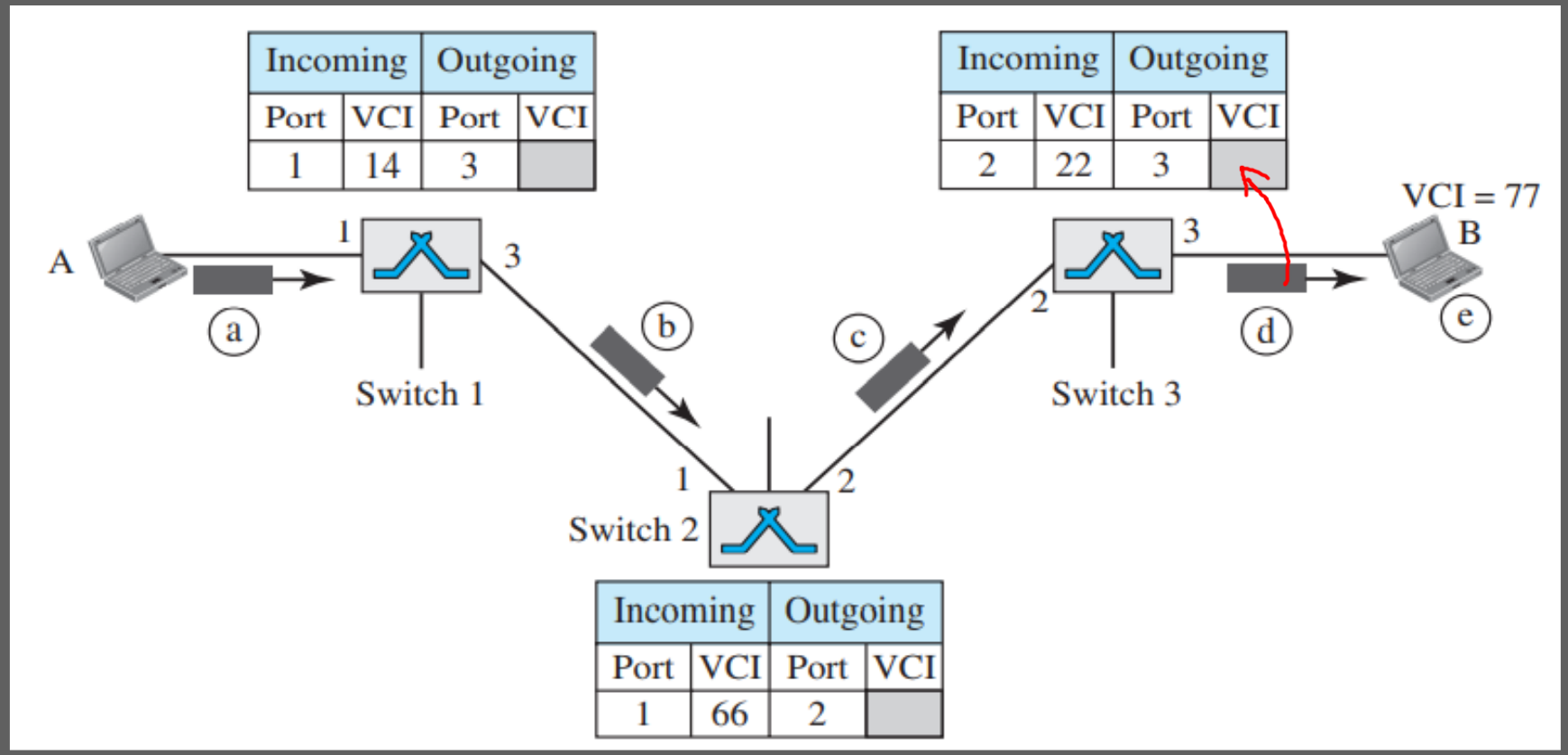
Setup Phase

- In the setup phase, a switch creates an entry for a virtual circuit.
- For example, suppose source A needs to create a virtual circuit to B.
- Two steps are required: the setup request and the acknowledgment.

Source-to-destination data transfer in a virtual-circuit network



Setup request in a virtual-circuit network



CONTD…

- **A.** Source A sends a setup frame to switch 1.
- **b.** Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3.
- The switch, in the setup phase, acts as a packet switch; it has a routing table which is different from the switching table.
- The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns.
- The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and outgoing port (3).

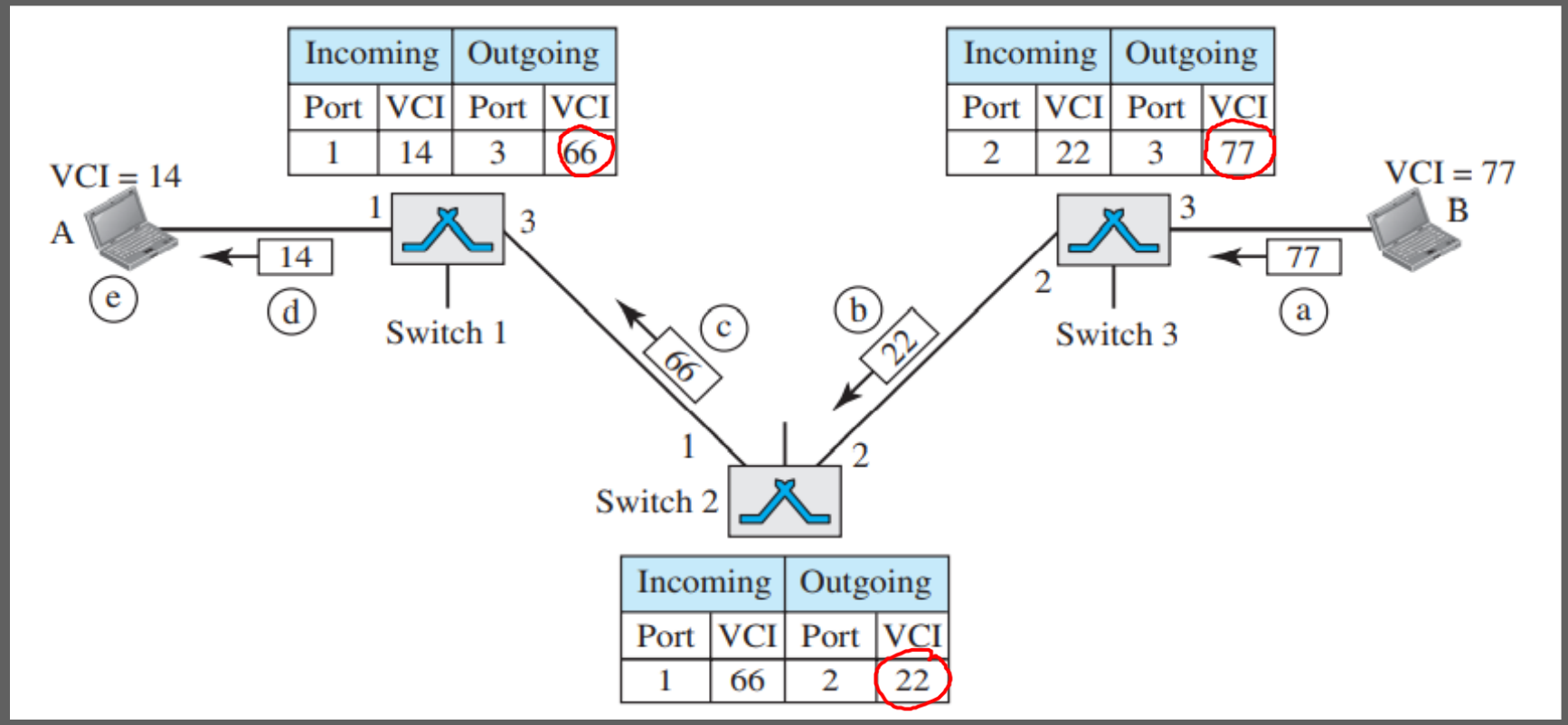
CONTD…

- The switch then forwards the frame through port 3 to switch 2.
- **C.** Switch 2 receives the setup request frame.
- The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).
- **D.** Switch 3 receives the setup request frame.
- Again, three columns are completed: incoming port (2), incoming VCI (22), and outgoing port (3).

CONTD…

- **E.** Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77.
- This VCI lets the destination know that the frames come from A, and not other sources.

Setup acknowledgment in a virtual-circuit network



CONTD…

- **A.** The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed.
- The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.
- **B.** Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.

CONTD…

- **c.** Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.
- **d.** Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.
- **e.** The source uses this as the outgoing VCI for the data frames to be sent to destination B

TEARDOWN PHASE

- In this phase, source A, after sending all frames to B, sends a special frame called a teardown request.
- Destination B responds with a teardown confirmation frame.
- All switches delete the corresponding entry from their tables

In virtual-circuit switching, all packets belonging to the same source and destination travel the same path, but the packets may arrive at the destination with different delays if resource allocation is on demand.

DELAY IN VIRTUAL-CIRCUIT NETWORKS

- In a virtual-circuit network, there is a one-time delay for setup and a one-time delay for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets.
- The packet is traveling through two switches (routers). There are three transmission times (3T), three propagation times (3τ), data transfer depicted by the sloping lines, a setup delay and a teardown delay.
- The total delay time is

Total delay = 3T + 3τ + setup delay + teardown delay

ACTIVITY